

HDOC Day-7 Activity

- Question 1:

Write a C Program to find frequency when angular frequency is given:

- **Step 1: Understanding the Problem**

- **Input:** Angular frequency ω in rad/s

- **Output:** Frequency f in Hz

- **Formula:**

- $\omega = 2\pi f$

Test Case	Input	Expected Output	Actual Output	Result
1	6.2832	1.00 Hz	1.00 Hz	Pass
2	12.5664	2.00 Hz	2.00 Hz	Pass
3	31.416	5.00 Hz	5.00 Hz	Pass

- C Programming

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float omega, frequency;
```

```
    const float PI = 3.14159;
```

```
    printf("Enter angular frequency (rad/s): ");
```

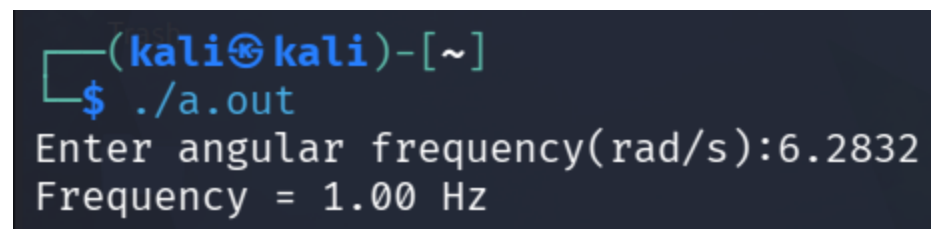
```
    scanf("%f", &omega);
```

```
frequency = omega / (2 * PI);  
printf("Frequency = %.2f Hz\n", frequency);  
  
return 0;  
}
```

Compilation and Execution:

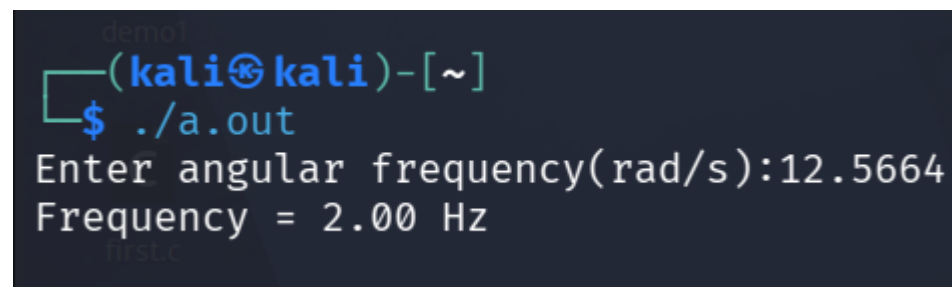
The code was successfully compiled and executed.

Test Case 1:

A terminal window with a dark background. The prompt is (kali@kali)-[~]. The user enters ./a.out. The program prompts for angular frequency, and the user enters 6.2832. The program outputs Frequency = 1.00 Hz.

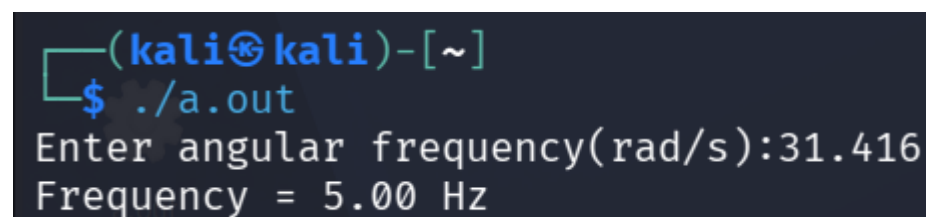
```
(kali@kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):6.2832  
Frequency = 1.00 Hz
```

Test Case 2:

A terminal window with a dark background. The prompt is (kali@kali)-[~]. The user enters ./a.out. The program prompts for angular frequency, and the user enters 12.5664. The program outputs Frequency = 2.00 Hz.

```
(kali@kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):12.5664  
Frequency = 2.00 Hz
```

Test Case 3:

A terminal window with a dark background. The prompt is (kali@kali)-[~]. The user enters ./a.out. The program prompts for angular frequency, and the user enters 31.416. The program outputs Frequency = 5.00 Hz.

```
(kali@kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):31.416  
Frequency = 5.00 Hz
```

Compilation and Execution Screenshot:

```
(kali㉿kali)-[~]  
$ vi remote.c  
  
(kali㉿kali)-[~]  
$ cc remote.c  
File System  
(kali㉿kali)-[~]  
$ .  
.: not enough arguments  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):6.2832  
Frequency = 1.00 Hz  
demo1  
(kali㉿kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):12.5664  
Frequency = 2.00 Hz  
in stc  
(kali㉿kali)-[~]  
$ ./a.out  
Enter angular frequency(rad/s):31.416  
Frequency = 5.00 Hz
```

- Question 2 :

Write a C Program to find Electric field between two parallel plates when voltage and separation are given.

- **Input:**
 - Voltage V in volts
 - Separation d in metres
- **Output:** Electric field E in V/m
- **Formula:** $E = V/d$

Test Case	Input	Expected Output	Actual Output	Result
1	100V ,0.5m	200 V/m	200.00 V/m	Pass
2	200V, 0.2m	1000 V/m	1000.00 V/m	Pass
3	500V,0.25m	2000 V/m	2000.00 V/m	Pass

- C Programming

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float voltage, separation, electric_field;
```

```
    printf("Enter voltage (V): ");
```

```
    scanf("%f", &voltage);
```

```
    printf("Enter separation between plates (m): ");
```

```
    scanf("%f", &separation);
```

```
electric_field = voltage / separation;  
  
printf("Electric Field = %.2f V/m\n", electric_field);  
  
return 0;  
}
```

Test 1:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):100  
Enter separation between plates (m): 0.5  
Electric Field = 200.000000 V/m
```

Test 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):200  
Enter separation between plates (m): 0.2  
Electric Field = 1000.000000 V/m
```

Test 3:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):500  
Enter separation between plates (m): 0.25  
Electric Field = 2000.000000 V/m
```

Compilation and Execution Screenshot:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):200  
Enter separation between plates (m): 0.2  
Electric Field = 1000.000000 V/m  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):500  
Enter separation between plates (m): 0.25  
Electric Field = 2000.000000 V/m  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter voltage (V):100  
Enter separation between plates (m): 0.5  
Electric Field = 200.000000 V/m
```

• Question 3 :

Write a C Program to find activity of a radioactive substance when decay constant and number of nuclei are given.

• **Step 1: Understanding the Problem**

• **Input:**

- Decay constant λ in s^{-1}
- Number of nuclei N

• **Output:** Activity A in Becquerel (Bq)

• **Formula:** $A = \lambda N$

Test Case	Input (λ, N)	Expected Output	Actual Output	Result
1	0.01, 1000	10 Bq	10.00 Bq	Pass
2	0.002, 5000	10 Bq	10 Bq	Pass
3	0.05, 2000	100 Bq	100 Bq	Pass

• C Programming

```
#include <stdio.h>
int main()
{
    float lambda, activity;
    long long N;
    printf("Enter decay constant (per second): ");
    scanf("%f", &lambda);

    printf("Enter number of nuclei: ");
    scanf("%lld", &N);

    activity = lambda * N;

    printf("Activity = %.2f Bq\n", activity);
    return 0;
}
```

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter decay constant (per second):0.01  
Enter number of nuclei: 1000  
Activity = 10.00 Bq
```

Test 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter decay constant (per second):0.002  
Enter number of nuclei: 5000  
Activity = 10.00 Bq
```

Test 3 :

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter decay constant (per second):0.05  
Enter number of nuclei: 2000  
Activity = 100.00 Bq
```

Compilation and Execution Screenshots:


```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter decay constant (per second):0.01

Enter number of nuclei: 1000

Activity = 10.00 Bq

first.c

```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter decay constant (per second):0.002

Enter number of nuclei: 5000

Activity = 10.00 Bq

```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter decay constant (per second):0.05

Enter number of nuclei: 2000

Activity = 100.00 Bq